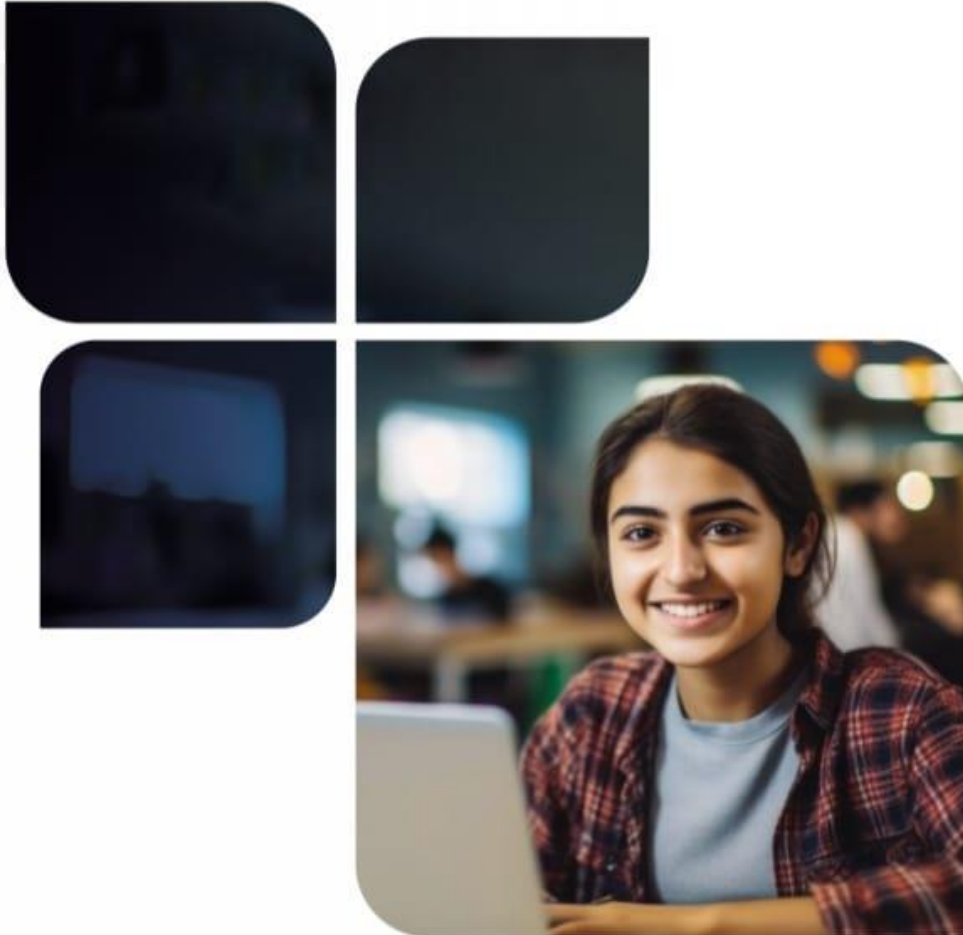


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PROJECT REPORT

TITLE : PLAGARISM CHECKER

Submitted by : P. Rajasree
Submitted to : Global Quest Technologies
Date : 13 June 2026

ABSTRACT

The **Plagiarism Checker** is a Python-based application designed to detect similarity between text documents and identify potential instances of copied content. The system compares input text files by analyzing words, phrases, and sentence structures to calculate a similarity score, helping users determine the originality of the content.

This project is developed using core and advanced Python concepts such as string processing, file handling, and basic text analysis techniques. The application reads multiple documents, preprocesses the text by removing unnecessary characters, and applies comparison methods such as word matching or frequency analysis to measure similarity. The result is presented as a percentage or score, indicating the level of plagiarism between the documents.

The Plagiarism Checker demonstrates the practical application of Python in text analysis and content verification. It can be further enhanced by integrating advanced techniques such as Natural Language Processing (NLP), semantic analysis, and support for multiple file formats like PDF and DOCX, making it a more robust and intelligent system for academic and professional use.

SOURCE CODE:

```
import re
from difflib import SequenceMatcher

# ♦ Step 1: Read and clean code
def read_and_clean(file_path):
    with open(file_path, 'r', encoding='utf-8') as file:
        code = file.read()

    # Remove comments
    code = re.sub(r'#. *', '', code)

    # Remove multi-line comments (""" """)
    code = re.sub(r'("""[\s\S]*?""")', '', code)

    # Remove extra whitespace
    code = '\n'.join(line.strip() for line in code.splitlines() if line.strip())

    return code

# ♦ Step 2: Calculate similarity
def calculate_similarity(code1, code2):
    return SequenceMatcher(None, code1, code2).ratio()

# ♦ Step 3: Line-by-line matching
def find_matching_lines(code1, code2):
    lines1 = code1.split('\n')
    lines2 = code2.split('\n')

    matches = []

    for i, line1 in enumerate(lines1):
        for j, line2 in enumerate(lines2):
            similarity = SequenceMatcher(None, line1, line2).ratio()
            if similarity > 0.8 and line1 != '':
                matches.append((i+1, j+1, line1))

    return matches

# ♦ Main Function
def plagiarism_checker(file1, file2):
    code1 = read_and_clean(file1)
    code2 = read_and_clean(file2)

    similarity = calculate_similarity(code1, code2) * 100

    print(f"\n🔍 Similarity Score: {similarity:.2f}%")

    if similarity > 80:
        print("🚩 High Plagiarism Detected!")
    elif similarity > 50:
        print("⚠️ Moderate Similarity Detected!")
    else:
        print("✅ Low Similarity")

    print("\n🔗 Matching Lines:")
    matches = find_matching_lines(code1, code2)

    for m in matches:
        print(f"File1 Line {m[0]} == File2 Line {m[1]} → {m[2]}")

# ♦ Run Program
if __name__ == "__main__":
    file1 = input("Enter first file path: ")
    file2 = input("Enter second file path: ")

    plagiarism_checker(file1, file2)
```

FILE1.PY

```
# Program to print numbers

def print_numbers():
    for i in range(10):
        print(i)

print_numbers()
```

FILE2.PY

```
# Printing numbers using loop

def display_numbers():
    for num in range(10):
        print(num)

display_numbers()
```

OUTPUT:-

```
PS C:\Users\chand\OneDrive\Pictures\Desktop\ggt projects> & c:\python313\python.exe "c:/Users/chand/OneDrive/Pictures/Desktop/ggt projects/plagiarism_checker.py"
Enter first file path: file1.py
Enter second file path: file2.py

🔍 Similarity Score: 79.71%
🚩 Moderate Similarity Detected!

💡 Matching Lines:
File1 Line 2 == File2 Line 2 → for i in range(10):
PS C:\Users\chand\OneDrive\Pictures\Desktop\ggt projects>
```



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#324, 2nd Floor, 3 A Cross, Near
Seshadripuram First Grade College,
Above City Union Bank,
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Bengaluru-560064

+91 9448 403 469 | 080-49720009
info@gqtech.in | www.gqtech.in